FACULTY OF ACCOUNTING AND INFORMATICS

DEPARTMENT OF INFORMATION TECHNOLOGY

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This declaration must be completed and submitted to your respective group lecturer for all phases of the project

Student No.	Student Initials & Surname	Signature
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1. We understand what plagiarism is and we are aware of the DUT'S policy in this regard.
2. We declare that this tutorial/project is our work.
3. Where other people's work has been used (either from a printed source, the internet, or any other source) this has been carefully acknowledged and referenced following departmental requirements.
4. We have not used other students' past work to hand in as our own.
5. We have not allowed and will not allow anyone to copy our work to pass it off as their work.

Group Name (E.g. Group A, Group H, Group AI): _____

Please complete this form and submit it as a cover sheet every time you submit your project. One submission is required per group. The details completed below must be correct.

Student No.	Surname, Initials	Group (App Dev, App Dev 4Yrs, or BA)	Task Completed	Signature
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Overview

The DUT Housing Department aims to enhance manual operations by implementing the Digital Residence Management System (DRMS). The DRMS will automate administrative tasks, improve security measures, and promote sustainability. It will manage student records, residence allocation and room assignments, provide an online portal for maintenance requests, and maintain a digital log for visitors. Transitioning from paper-based processes to the DRMS is expected to increase operational efficiency, enable faster staff responses to student needs, generate actionable data, and advance the university's sustainability objectives. These improvements aim to establish a safer and more efficient residential environment for all stakeholders.

New Processes

With the new processes, the new system allows students to register within the web application, allowing them to enter their details, which include (First Name, Last Name, Email address, phone number, password, student number, faculty, and year of study)

The same processes go for the Staff/ Admin (First Name, Last Name, Email address, phone number, password, staff number, role)

After the registration process, each user can log in to their assigned dashboard. For the student, they access the student residence management dashboard and view maintenance requests, active announcements, pending requests, and their allocated residence status.

The student can report and track maintenance issues, view their profile, and see their assigned residence and download proof of residence registration.

An admin can view the students who have registered within the system, register and view the total number of residences that are within the system, can send announcement to residence manager and students and allocate registered students to residences according to their selected faculty and manage user within the system which include the students and building manager.

The building manager can view the total number of students within that residence that he or she owns, send announcements to students, view maintenance reports and mark them as resolved once they are resolved, and he/ she can also view announcements coming from the administration.

Future processes

For our next phase, we will focus on allocating the students to their rooms. Students can sign in visitor the system will track their arrival time and time out.

The student can also add an image when reporting a maintenance issue.

A QR Code will redirect the user to their profile page and allow a user to download proof of registration. When an admin is sending out announcements, a new feature will be added that allows them to use voice recognition.

Problem description and challenges

DUT Housing Department ensures that students are allocated to their residences, resolve issues that may arise, they also ensure that student comply with their policies. Current processes include:

- Manual Applications: Students apply for residences through email and are allocated based on faculty or random assignment.

- Physical Room Allocation: Students stand in long queues and submit hard copies of proof of residence for verification.

- Maintenance Reporting: Issues are reported in WhatsApp groups or handwritten logbooks, which are reviewed only weekly.

- Visitor Registration: Visitors are registered manually in a physical book, making tracking inefficient. Key Challenges:

- Long queues and delays in room allocation. - Risk of data loss or misinterpretation due to hard copy records. - Slow response to maintenance issues as logs are checked weekly. - Lack of real-time monitoring of visitors and residence occupancy

Proposed solution and system capabilities

The proposed solution is a web-based system that digitizes and integrates all core residence management activities into a single platform

System capabilities

1. Student Records Management

- Stores student credentials and residence allocation details in a secure database.

- Provides real-time access to records for administrators.

2. Online Registration & Residence Allocation

- Students register online for residence

- Students are allocated automatically based on availability, faculty.

. - Eliminates physical queues through digital room allocation.

3. Maintenance Management

- Allows students to report maintenance issues via the platform.

- Tracks resolution progress and timelines for quick response.

4. Visitor Management

- Digital registration of visitors with ID verification (Student number, ID number).

- Tracks visitor entry and exit times to ensure visiting policies are not violated.

6. Green Computing & Data Security

- Reduces paper usage through complete digitization.

- Implements encryption and secure authentication for data protection.

Entities

Residence - Includes residence details, room availability, and assigned students.

Room - Tracks room status, occupancy, and allocation history.

Student - Contains student credentials, residence status, complaints, and room allocation.

Admin - Housing staff responsible for managing allocations, complaints, and oversight.

Visitors - Contains visitor credentials, check-in, and check-out times.

Maintenance - Logs reported issues, status updates, and resolution timelines.

Announcement- contains broadcast messages from the administration or residence building manager, including titles, content, priorities, and target audiences.

Benefits

1. Efficiency: Eliminates manual processes, saving time for students and staff.

2. Data Security: Protects student records through secure digital storage.

3. Improved Communication: Real-time updates on maintenance and housing notices.

4. Sustainability: Reduces paper use and promotes eco-friendly practices.

5. Compliance: Enforces visitor and residence policies through automated checks.

6. Promotes green computing and saves money: By making all processes digital, a lot of money for buying books is saved, and less paper is required.

Technology stacks

Web dashboard- ASP.NET MVC (C#)

Database- SQL Server

API/Backend- ASP.NET Web API

Authentication- Dut student email login

UI/UX Design- Razor views for web

Implementation Plan

Phase 1: Information Gathering & System Design

- Conduct detailed analysis with the housing department and collect information.

- Develop system plan and design mock up.

Phase 2: Database Integration

- Build central database that stores student registration, room allocation, maintenance, announcements, and visitor records.

Phase 3: Testing & Pilot Deployment

- Test system with a single selected residence.

- Collect feedback and optimize performance.

Phase 4: Full Deployment & Training

- Deploy system to the selected number of residences.

- Create a manual guide that explains how to use the system.

- Host training sessions with staff and students to show them how to use the system.

Business Rules

1. Eligibility & Application Rules

1.1 Academic Standing

- A student must be fully a registered student for the current academic year/semester to be eligible to apply for residence.

1.2 Financial Standing

- A student must have no outstanding tuition fees from previous semesters to be eligible to apply.

- A student must be coded by NSFAS for allocation to take place.

1.3 First-Year Priority

- A certain percentage of rooms (70%) are reserved for first-year students during the main allocation period.

1.3.1 Application Window

- Applications for residence must be submitted within the officially announced dates. The system must reject any applications submitted after the deadline.

1.4 Single Application

- A student may only have one active application per academic semester/year.

2. Allocation & Assignment Rules

2.1 Allocation Priority Order

2.1.1 Students must be allocated based on the following priority order:

- Students with documented disabilities (requiring specific accommodation).
- First-year students coming from outside Durban CBD and neighbouring area
- Date of application submission (first-come, first-served).

2.2 Gender Separation

- Students can only be assigned to rooms in residences designated for their gender.

2.3 Room Capacity

- A room cannot be assigned to more students than its designated capacity. e.g., a “Double Room" must have exactly two students.

2.4 Conditional Assignment

- A room assignment is conditional until full payment of residence fees is confirmed.

3. Financial Rules

3.1 Fee Structure

- Residence fees are determined by the room type and residence hall. The system must calculate the correct fee based on the student's allocation.

3.2 Payment Confirmation

- The system must automatically update a student's allocation status to "Confirmed" only upon successful verification of full payment from the payment gateway or finance office.

4. Maintenance & Operations Rules

4.1 Issue Categorization & Priority

- Maintenance requests must be categorized and prioritized automatically.

-Critical Response within 24h: Issues affecting health/safety e.g. electrical hazards, major leaks, no water.

-Urgent Response within 72h: Issues affecting quality of life e.g. broken window, malfunctioning lock, no power.

-Routine Response within 7 days: Cosmetic or non-critical repairs e.g. peeling paint, broken drawer.

4.2 Request Ownership

- Only the currently assigned occupants of a room can submit maintenance requests for that room.

5. General & System Rules

5.1 Data Integration

- Student personal data (Name, ID, Program) must be sourced from the DUT admission office and cannot be edited within the residence system.

5.2 Role-Based Access

- Administrators have access based on their role:

-Finance Officer: Can view payment statuses and generate financial reports. Cannot allocate rooms.

-Maintenance Manager: Can view and assign maintenance tickets. Cannot access financial data.

-Residence Administrator: Can manage applications, allocations, and announcements. Cannot modify financial records.

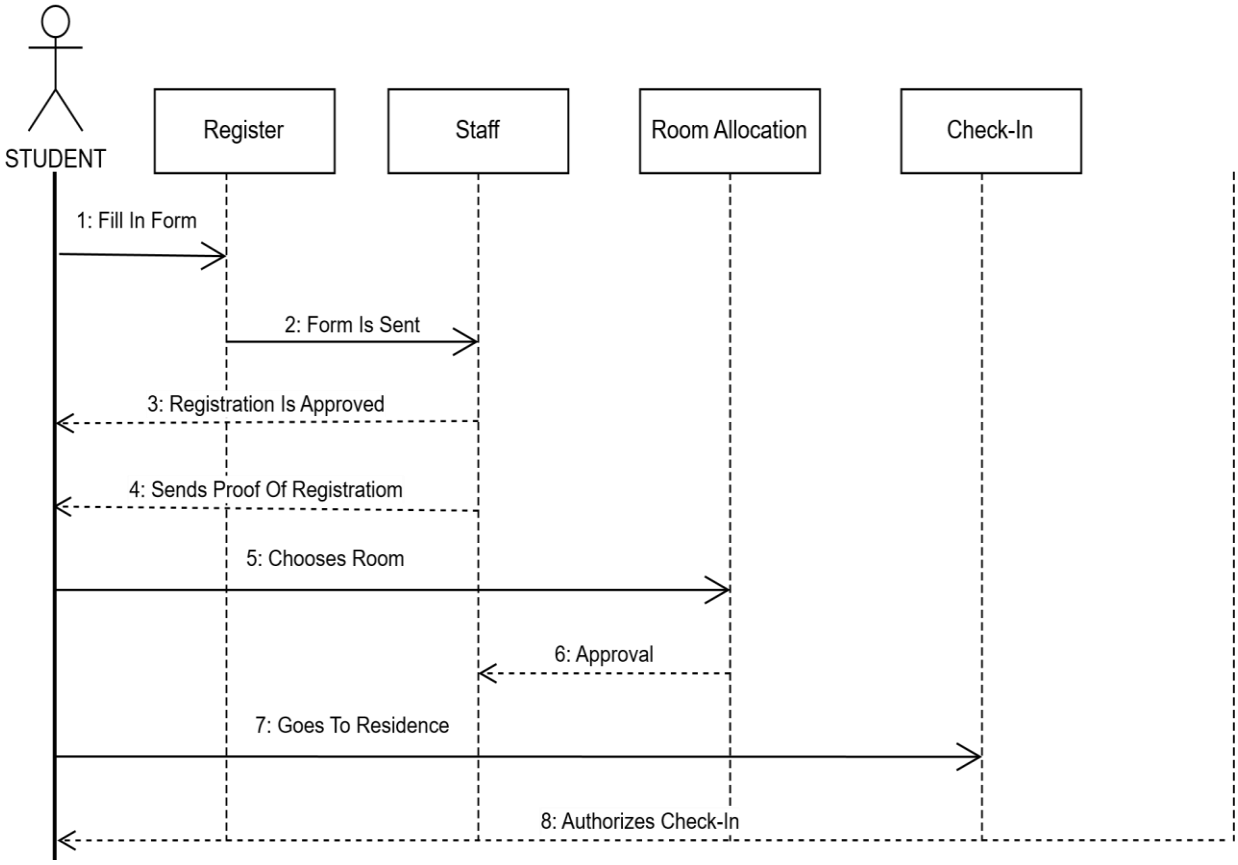
5.3 Data Retention

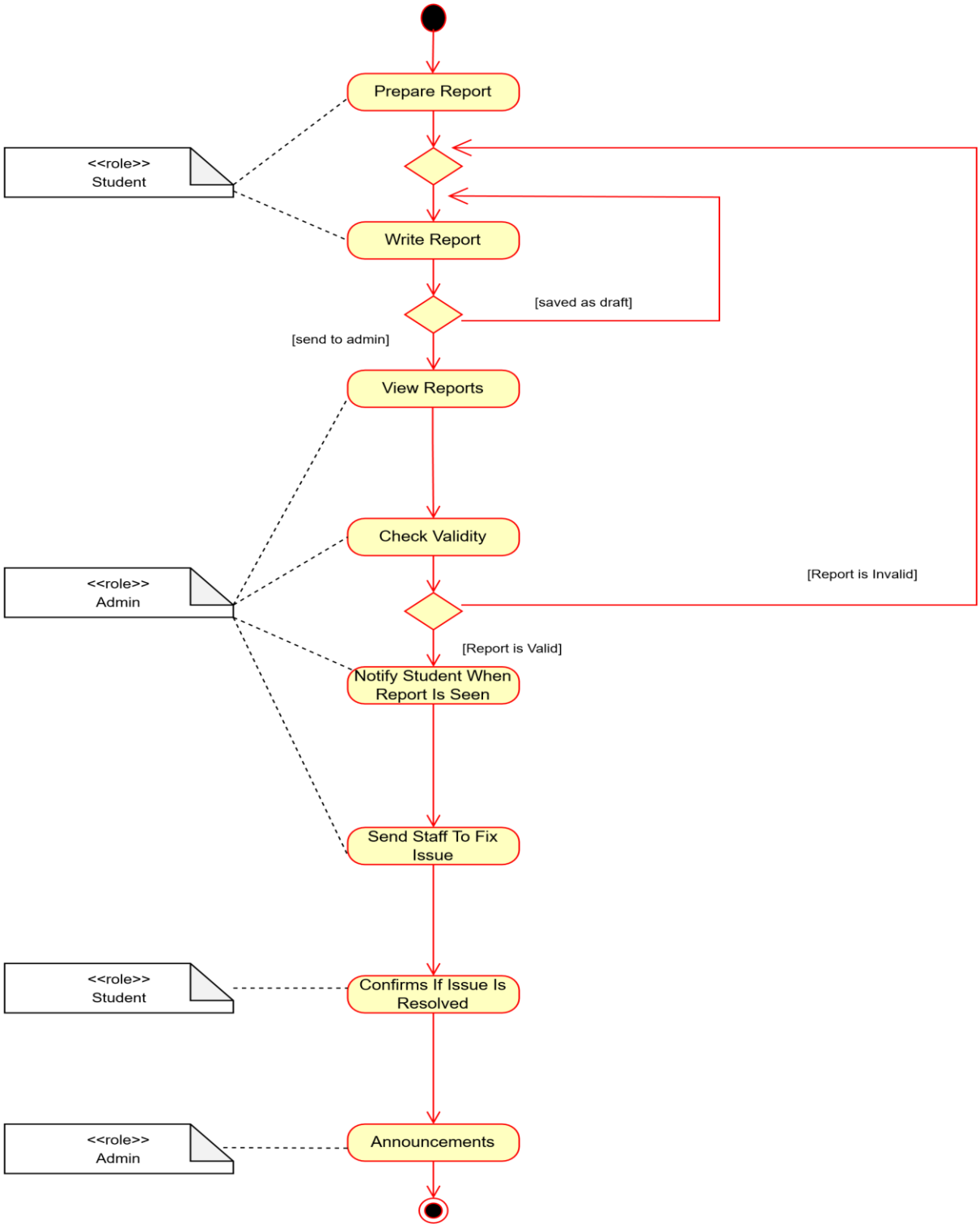
- All application, allocation, and transaction records must be archived in the system database for a minimum of five years after a student graduates or leaves the university.

CONCLUSION

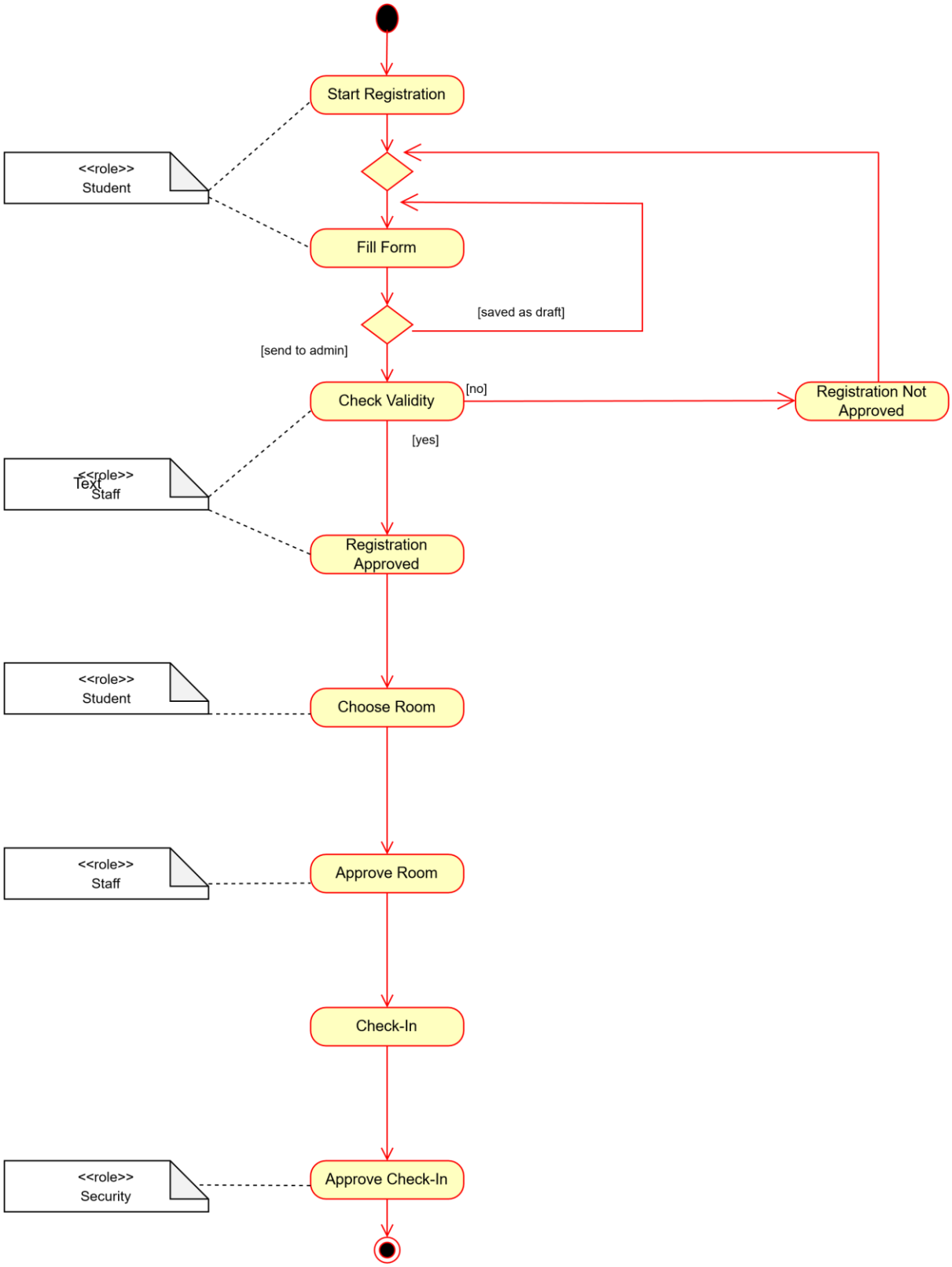
The proposed System will be of significant help to the Housing department since it will ensure that a lot of money is by reducing the number of books the company buys since all records will be stored online, this will also promote green computing. Not only will it save money, but it will also save time since all processes will be automated.

Maintenance Activity Diagram





Registration Activity Diagram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Fill Form
[saved as draft]
[send to admin]
Check Validity
[no]
Registration Not Approved
[yes]
<<role>>
Staff
Text
Registration Approved
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Student
Choose Room
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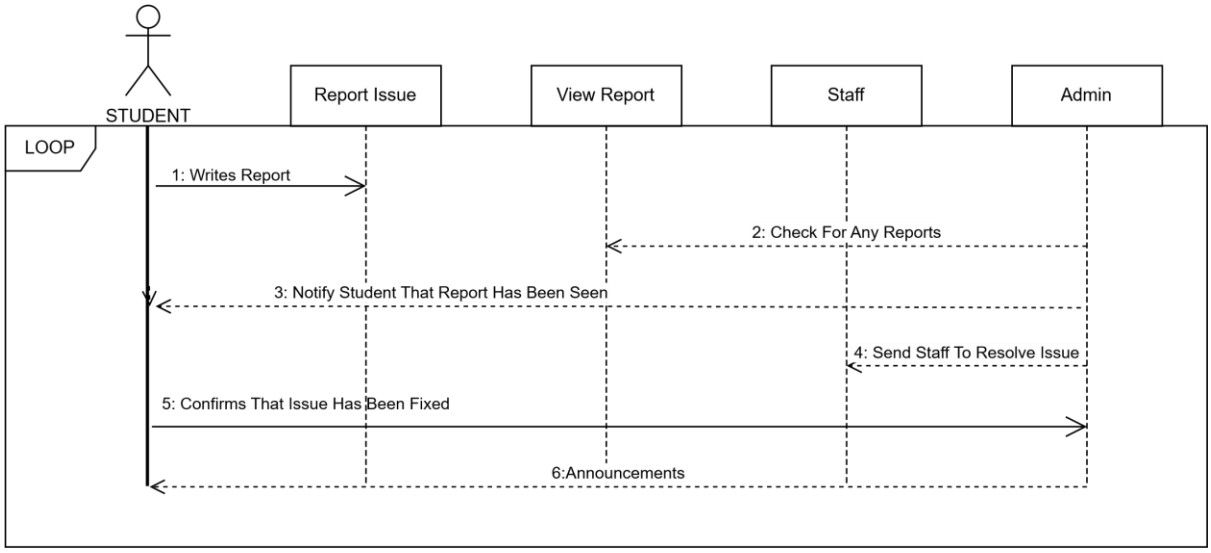
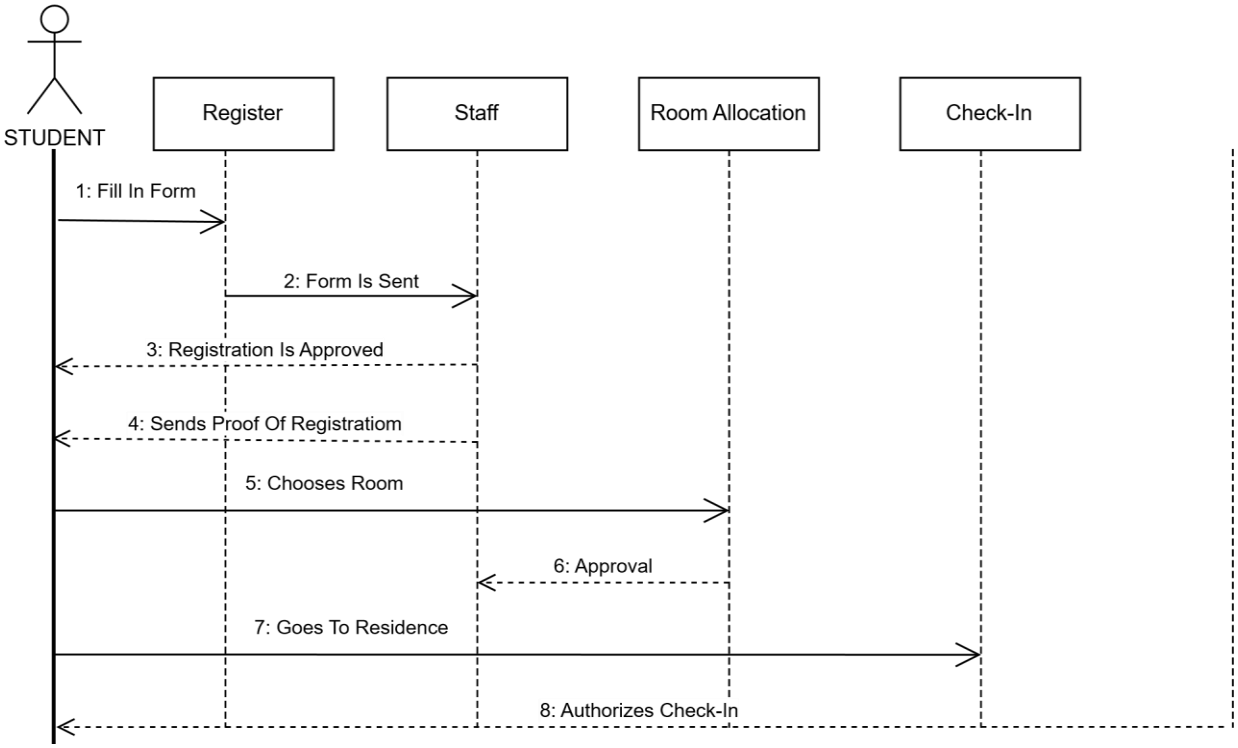
Staff
Approve Room
Check-In
<<role>>
Security
Approve Check-In

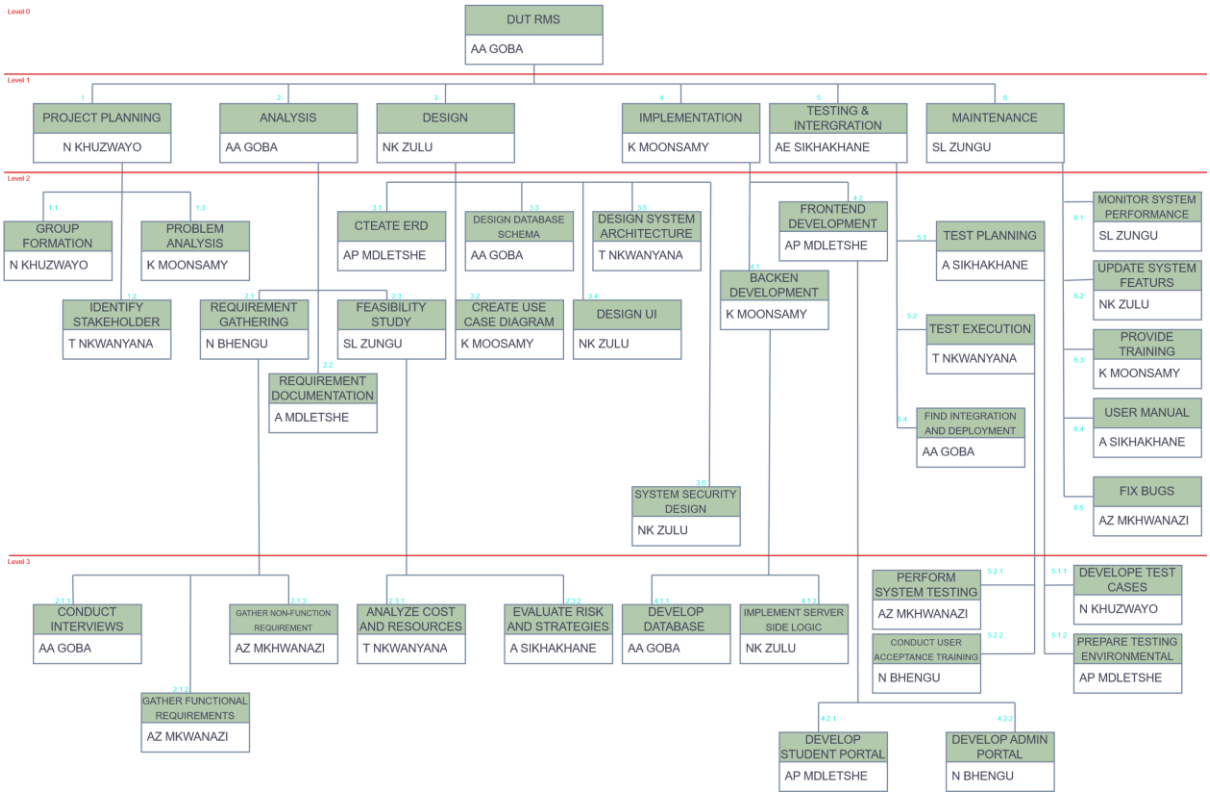
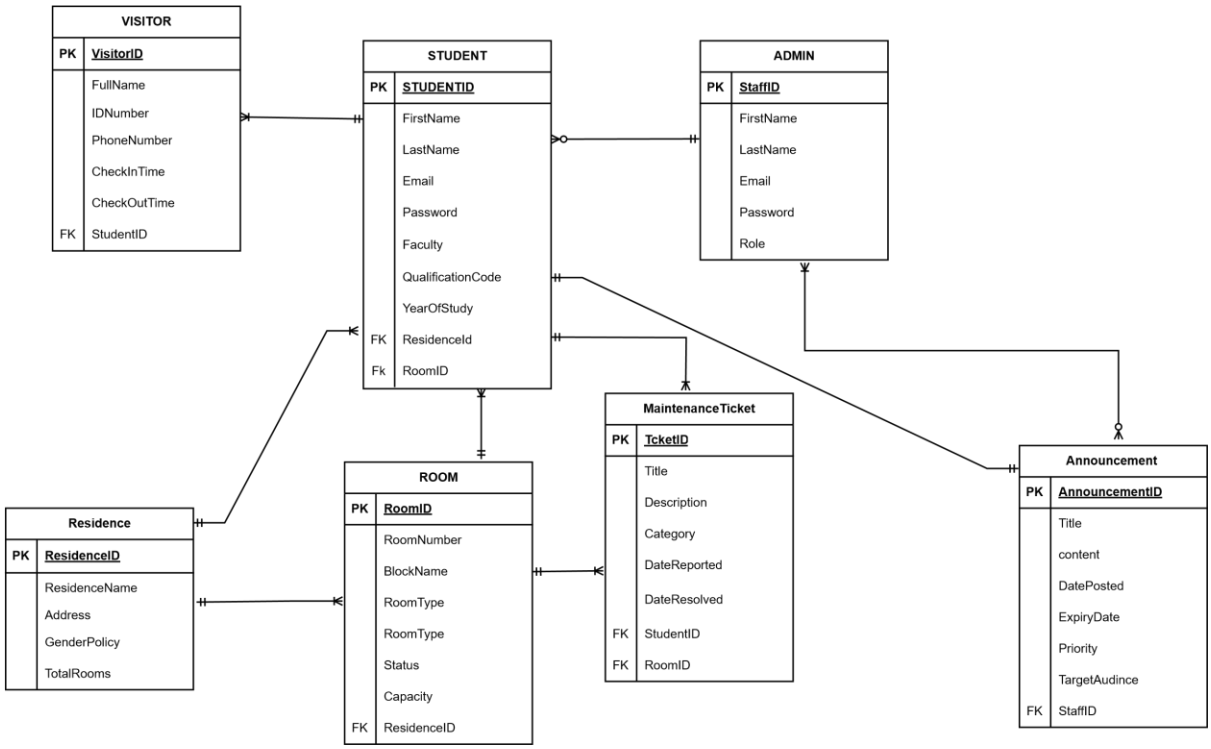
Sequence Diagram





WBS ITEMS	Work items	KHUZWAYO	GOBA	ZULU	MOONSAMY	ZUNGU	SIKHAKHANE	MDLETSHE	BHENGU	MKHWANAZI	NKWANYANA
1	**Project Planning**	**P**		S			S	S			
1.1	Group Formation	**P**				S					
1.2	Identify stakeholder									S **P**	
1.3	Problem analysis		S		**P**						
2	**Analysis**		**P**								
2.1	Requirements gathering				S				**P**		
2.1.1	Conduct Interviews with housing		**P**			S	S	S	S		
2.1.2	Gather functional requirements	S		S						**P** S	
2.1.3	Gather non-functional requirements		S		S				S	**P**	
2.2	Requirements documentation	S				S		**P**			S
2.3	Feasibility study					**P**					
2.3.1	Analyze cost and resources			S			S		S		**P**
2.3.2	Evaluate risk and strategies		S		S		**P**				
3	**Design**			**P**							
3.1	Create ERD			S				**P**			
3.2	Create use case diagram				**P**	S					
3.3	Design database schema		**P**		S				S		
3.4	Design user interface			**P**	S						
3.5	Design system architecture									S **P**	
3.6	System security design			**P**		S		S			
4	**Implementation**				**P**						
4.1	Backen development		S	S	**P**		S				S

4.1.1	Develop database	**P**	S			
4.1.2	Implement server-side logic		**P**	S	S	
4.2	Fronted development	S		**P**	S S	S
4.2.1	Develop student portal			S	**P** S	
4.2.2	Develop admin portal		S		S **P**	
5	**Testing and integration**		S	**P**		S
5.1	Develop test case	**P**	S			S
5.1.1	Prepare testing environment		**P** S			S
5.1.2	Perform all system test			**P**	S	S
5.1.3	Conduct user acceptance test	S	S		**P**	
5.2	Integrate all system modules		S	S	S	**P**
5.3	Conduct training	**P**	S	S		S
6	**Maintenance**			**P**		
6.1	Monitor system performance	S	**P**		S	
6.2	Update system features			**P**		
6.3	Handover user manuals		S	S	**P**	
6.4	Fix bugs		S		S	**P**


Level 0
DUT RMS
AA GOBA
Level 1
1 PROJECT PLANNING
N KHUZWAYO
2 ANALYSIS
AA GOBA
3 DESIGN
NK ZULU
4 IMPLEMENTATION
K MOONSAMY
5 TESTING & INTERGRATION
AE SIKHAKHANE
6 MAINTENANCE
SL ZUNGU
Level 2
1.1 GROUP FORMATION
N KHUZWAYO
1.3 PROBLEM ANALYSIS
K MOONSAMY
1.2 IDENTIFY STAKEHOLDER
T NKWANYANA
2.1 REQUIREMENT GATHERING
N BHENGU
2.2 REQUIREMENT DOCUMENTATION
A MDLETSHE
2.3 FEASIBILITY STUDY
SL ZUNGU
3.1 CTEATE ERD
AP MDLETSHE
3.2 CREATE USE CASE DIAGRAM
K MOOSAMY
3.3 DESIGN DATABASE SCHEMA
AA GOBA
3.4 DESIGN UI
NK ZULU
3.5 DESIGN SYSTEM ARCHITECTURE
T NKWANYANA
3.6 SYSTEM SECURITY DESIGN
NK ZULU
4.1 BACKEN DEVELOPMENT
K MOONSAMY
4.2 FRONTEND DEVELOPMENT
AP MDLETSHE
5.1 TEST PLANNING
A SIKHAKHANE
5.2 TEST EXECUTION
T NKWANYANA
5.4 FIND INTEGRATION AND DEPLOYMENT
AA GOBA
6.1 MONITOR SYSTEM PERFORMANCE
SL ZUNGU
6.2 UPDATE SYSTEM FEATURS
NK ZULU
6.3 PROVIDE TRAINING
K MOONSAMY
6.4 USER MANUAL
A SIKHAKHANE
6.5 FIX BUGS
AZ MKHWANAZI
Level 3
2.0.1 CONDUCT INTERVIEWS
AA GOBA
2.1.2 GATHER FUNCTIONAL REQUIREMENTS
AZ MKWANAZI
2.1.3 GATHER NON-FUNCTION REQUIREMENT
AZ MKHWANAZI
2.3.1 ANALYZE COST AND RESOURCES
T NKWANYANA
2.3.2 EVALUATE RISK AND STRATEGIES
A SIKHAKHANE
4.1.1 DEVELOP DATABASE
AA GOBA
4.1.2 IMPLEMENT SERVER SIDE LOGIC
NK ZULU
5.2.1 PERFORM SYSTEM TESTING
AZ MKHWANAZI
5.2.2 CONDUCT USER ACCEPTANCE TRAINING
N BHENGU
5.1.1 DEVELOPE TEST CASES
N KHUZWAYO
5.1.2 PREPARE TESTING ENVIRONMENTAL
AP MDLETSHE
4.2.1 DEVELOP STUDENT PORTAL
AP MDLETSHE
4.2.2 DEVELOP ADMIN PORTAL
N BHENGU

Entity Relationship Diagram